Group Project Topic Plan

Project Title: Medical Appointment Attendance Prediction and Patient Behavior Pattern Research

Group Members

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Project Overview

Clean, analyze and mine a Medical Appointment Scheduling System dataset to uncover patient behavior insights and predict attendance, aiming to boost healthcare scheduling efficiency.

Research Objectives

1. **Predictive Analysis (Supervised Learning)**
   - Build models (Logistic Regression, Random Forest, XGBoost) to predict attendance (No-show vs Attended).
   - Identify key factors such as age, appointment gap, chronic conditions, and historical attendance.
2. **Pattern Discovery & Clustering (Unsupervised Learning)**
   - Use K-Means, DBSCAN to find patient behavior patterns.
   - Interpret clusters for targeted reminders or scheduling optimization.

Planned Methods & Tools

Phase	Description	Tools
Data Cleaning	Handle missing values, feature engineering	Python, Pandas
EDA & Visualization	Explore distributions and correlations	Matplotlib, Seaborn
Predictive Modeling	Train and evaluate models	scikit-learn, XGBoost
Clustering Analysis	Identify behavioral patterns	K-Means, DBSCAN
Reporting	Summarize findings and insights	Jupyter Notebook, Markdown

Expected Outcomes

- Predictive model with measurable accuracy for appointment attendance
- Identification of distinct patient clusters
- Improved scheduling efficiency and reduced no-show rates